3.5.1 Evolution of second-generation UAV development

The post-World War II era catalyzed significant research into UAVs, as global powers sought safer, more efficient ways to collect intelligence without risking pilots. Early developments focused on enhancing flight control systems, endurance, and payload capabilities, resulting in the first operationally effective UAVs. The Cold War became a defining period for UAV innovation, as both the United States and the Soviet Union invested heavily in aerial reconnaissance technologies.

By the 1960s and 1970s, the integration of jet propulsion, radio telemetry, and high-resolution imaging systems transformed UAVs into indispensable tools for intelligence gathering. The Vietnam War accelerated this trend, as UAVs demonstrated their ability to conduct real-time reconnaissance, damage assessment, and electronic warfare missions. Simultaneously, nations such as Israel began pioneering smaller, more efficient UAVs designed for real-time video transmission and battlefield coordination.

The 1980s and 1990s marked the refinement and operational maturity of UAVs through the incorporation of GPS navigation, digital flight controllers, and telemetry systems. This period also saw a diversification of drone applications, including civilian uses in mapping, meteorology, and environmental monitoring, establishing UAVs as multi-domain platforms.

3.5.2 Key pioneers and notable drone models

3.5.2.1 The Ryan Firebee (1951)

The Ryan Firebee (1951), developed by Ryan Aeronautical Company (United States), stands as one of the most significant milestones in the history of second-generation unmanned aerial vehicles (UAVs). Originally designed as a jet-powered target drone for the US Air Force and Navy, the